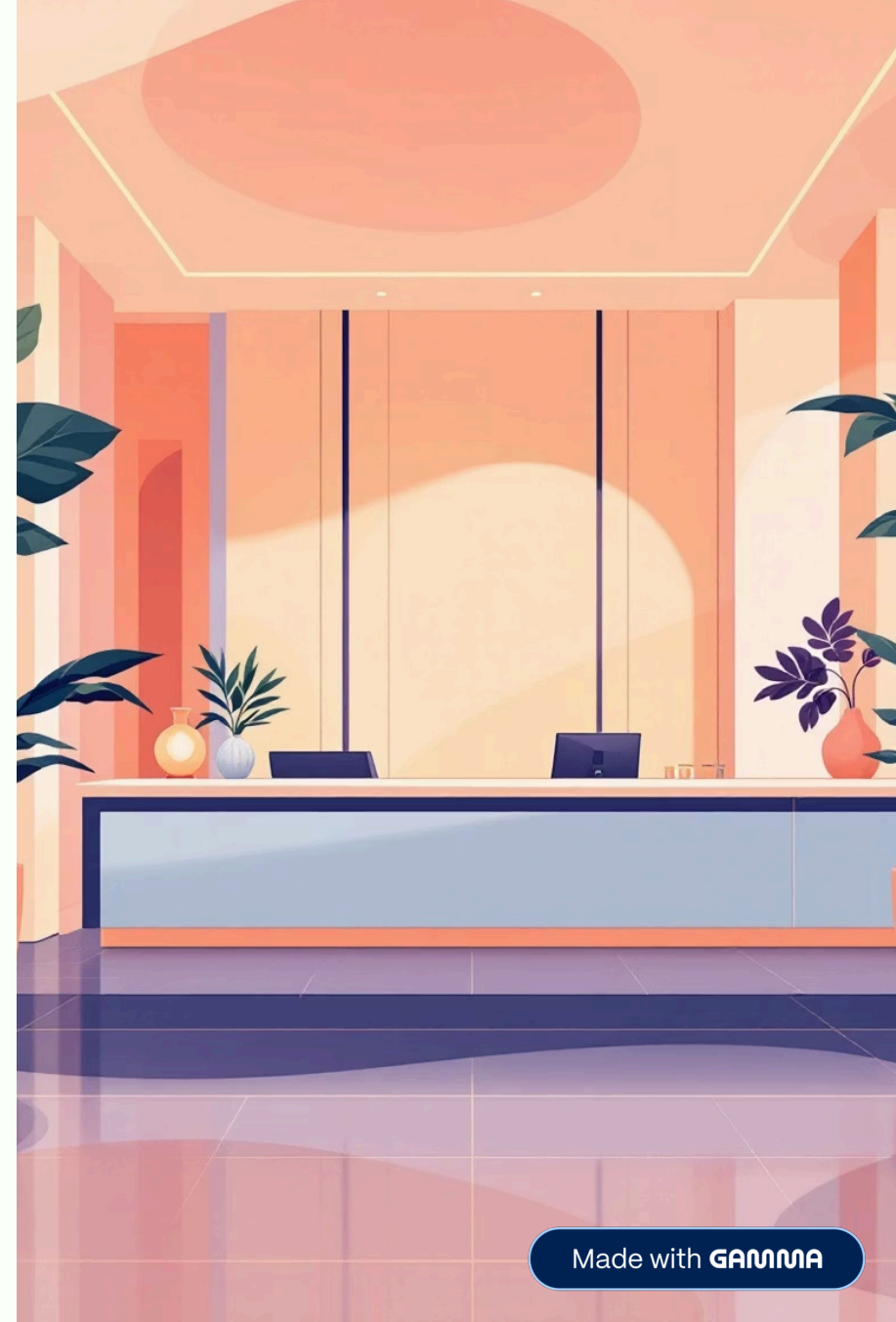


Hotel Booking Cancellations Analysis

Understanding the patterns behind booking cancellations to optimize revenue management and enhance customer satisfaction.



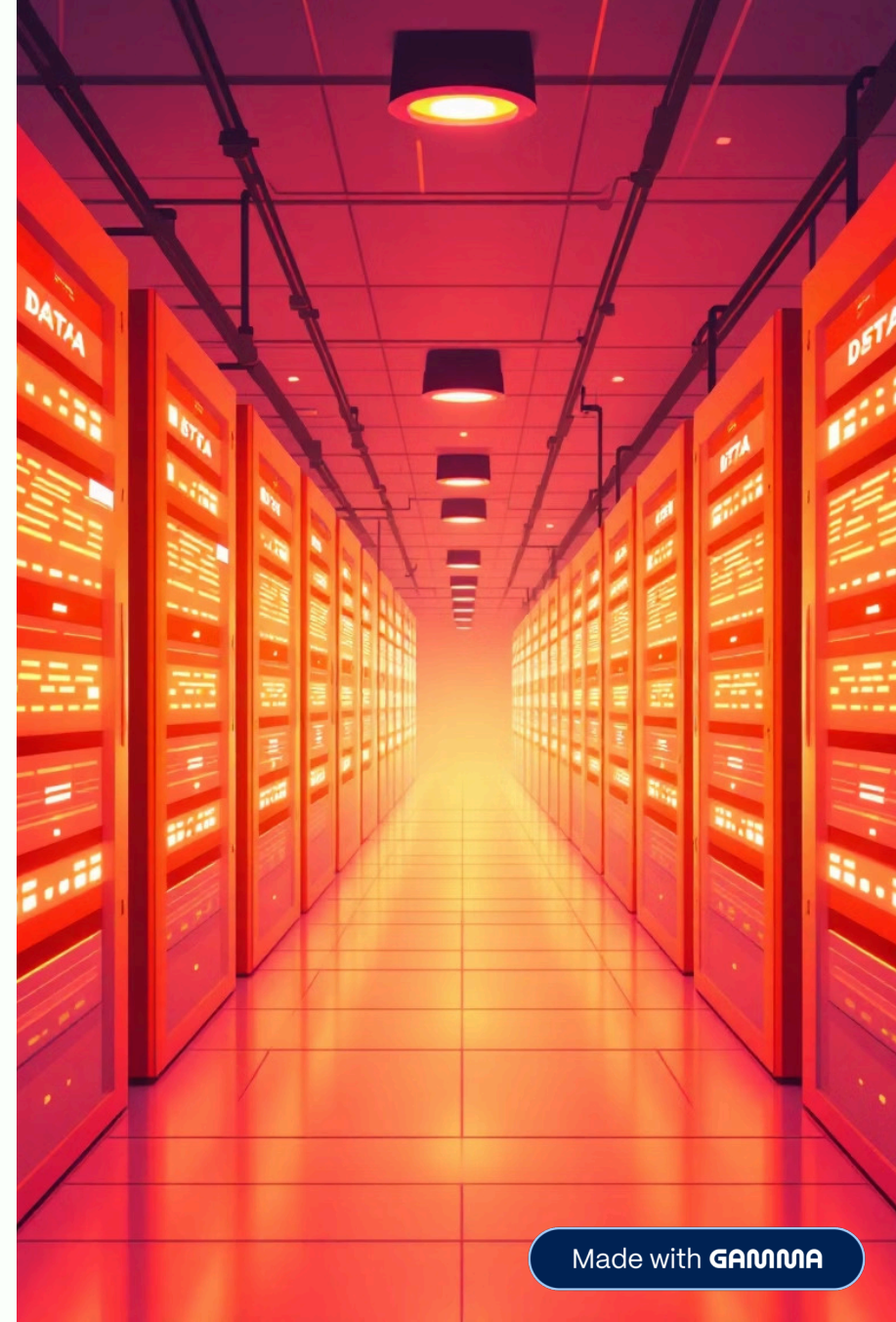
Project Objective

Analyze hotel booking data to identify key factors influencing cancellations. By understanding customer behavior, seasonal trends, and cancellation patterns, hotels can improve revenue management, reduce cancellations, and enhance customer satisfaction through data-driven strategies.



Dataset Overview

Comprehensive hotel booking records including customer demographics, booking details, lead times, room types, special requests, and cancellation status. Each record represents a single booking, providing insights into customer behavior and booking trends across multiple properties.



Tools & Methodology

Microsoft Excel

Data cleaning, pivot tables, and visualization

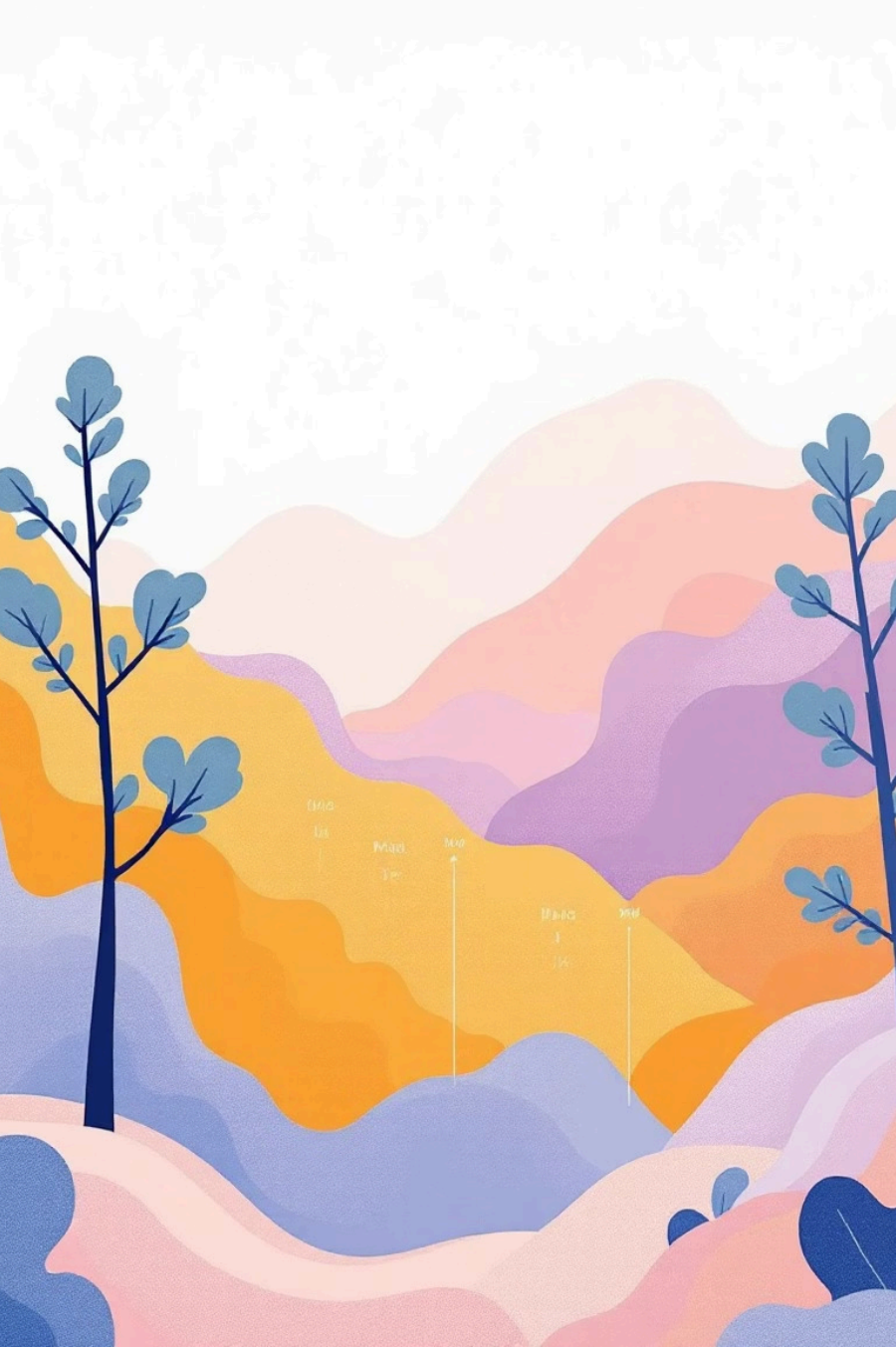
Python

Pandas, Matplotlib, Seaborn for analysis and trend identification

Power BI

Dashboard creation and interactive reporting





Lead Time Impact

A significant percentage of cancellations were made by customers with longer lead times. Bookings made far in advance show higher cancellation rates, suggesting customers with extended planning horizons are more likely to change plans.

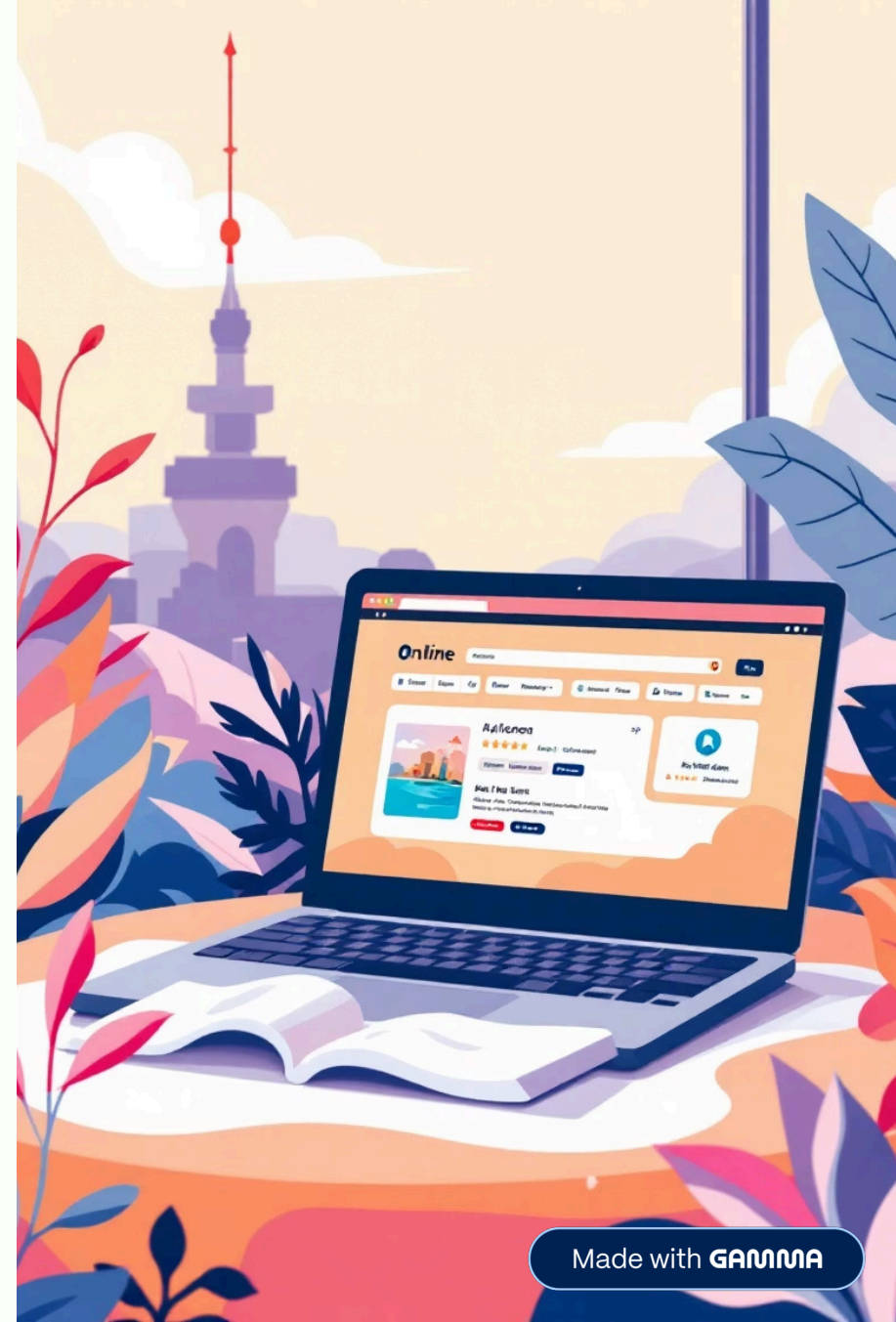
Booking Channel Analysis

Online Travel Agencies

Higher cancellation rates compared to direct bookings, indicating less customer commitment through third-party platforms.

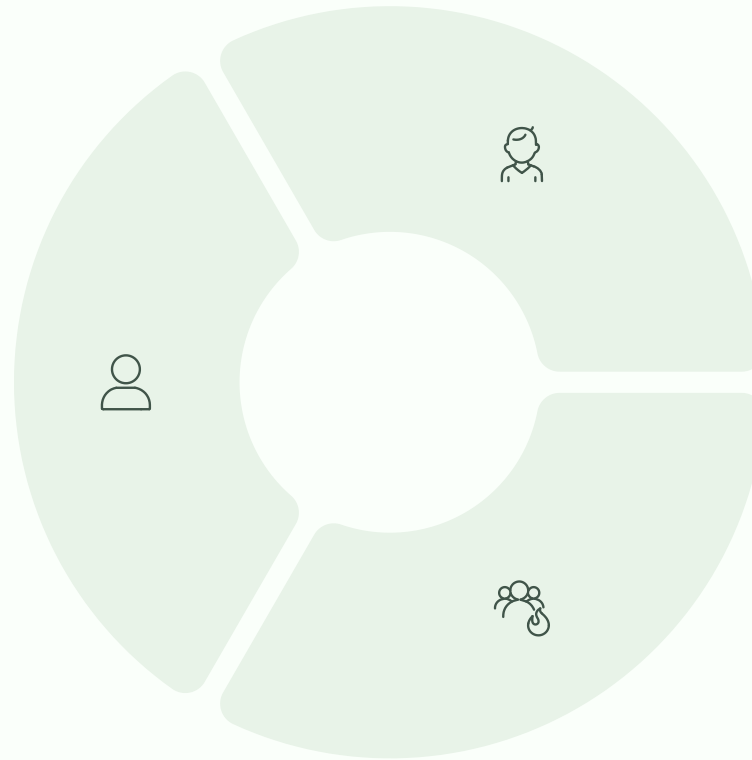
Direct Bookings

Lower cancellation rates, showing stronger customer commitment when booking directly with the hotel.



Customer Segment Patterns

Single Travelers
Higher cancellation rates



Families
Lower cancellation rates

Group Travelers
Lower cancellation rates



Seasonal & Pricing Dynamics

City hotels experienced more cancellations during peak season due to higher pricing. Customers booking at premium rates during high-demand periods showed greater flexibility to cancel, while off-season bookings demonstrated stronger commitment.



Special Requests Drive Commitment

Customers with special requests were significantly less likely to cancel, indicating higher commitment to their bookings. This suggests that personalized engagement and customization strengthen customer intent and reduce cancellation risk.



Strategic Recommendations

1 Adjust Pricing Strategies

Implement dynamic pricing to reduce peak-season cancellations while maintaining revenue.

2 Enhance Flexible Policies

Offer flexible cancellation terms to increase direct bookings and reduce OTA cancellations.

3 Personalize Customer Engagement

Encourage special requests and customization to build commitment and reduce cancellation rates.